

2. Requirement Analysis

Functional Requirements

- * Upload and read the dataset.
- * Perform preprocessing.
- * Handle missing values.
- * Train a Linear Regression model.
- * Save the trained model.
- * Accept user inputs through a web form.
- * Predict Human Development Index.
- * Display prediction results.

Non-Functional Requirements

- * User-friendly interface.
- * Fast prediction response.
- * Reliable prediction accuracy.
- * Easy maintenance.
- * Scalability.

Software Requirements

- * Python 3.x
- * Visual Studio Code
- * Flask
- * NumPy
- * Pandas
- * Matplotlib
- * Seaborn

- * Scikit-learn

- * Pickle

Hardware Requirements

- * Intel Core i3 or higher

- * 4 GB RAM or above

- * Windows/Linux/macOS

- * 500 MB free disk space

Dataset Requirements

- * HDI Dataset

- * CSV Format

- * Kaggle/Open Data Source

Expected Output

Prediction of Human Development Index based on user-provided socio-economic indicators.